

Why 1/N is a serious benchmark rather than a trivial benchmark

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The main problem lies in the fact that, while mean-variance portfolios are optimized for ideal diversification, this optimal diversification is undermined by estimation error when the portfolios are tested out-of-sample (OOS).

0.1 What information does Markowitz require?

The Markowitz model requires estimates of expected returns and the covariance matrix. In a traditional sample-based implementation, these quantities are estimated from historical returns, making the resulting portfolio sensitive to estimation error.

0.2 What information does the 1/N strategy require?

The 1/N strategy requires only the total capital to be deployed and the number of assets for investment, therefore, it does not estimate expected returns or covariances in order to determine portfolio weights. Consequently, its weights are not exposed to parameter-estimation error in $\hat{\mu}$ or $\hat{\Sigma}$

0.3 Where does estimation error come from?

Estimation error arises from market noise and the limited number of available observations.

Calculations were performed to determine the estimation window required to outperform the 1/N portfolio within the US stock universe. The results showed that for a portfolio of just 25 stocks, a 3,000-month window would be needed, but with traditional mean-variance portfolios typically use estimation windows of only 60 to 120 months.

0.4 Why can an in-sample optimal solution perform worse out-of-sample (OOS)?

Because the in-sample approach utilizes known market data, specifically the data for the assets being optimized. In the context of a mean-variance portfolio, the returns of the constituent stocks during a specific time window are used to calculate the expected return, essentially, no actual estimation is taking place.

In contrast, the OOS approach uses data up to a certain point in time to attempt to estimate the future expected performance of the stocks.

0.5 Why does this justify including 1/N in the research on the Mexican market?

The introduction mentions how various researchers have attempted different approaches to solving the estimation problem, such as using shrinkage estimators, imposing no-short-selling constraints, estimating probability regions for expected returns and optimizing based on them, and so on.

Across the seven empirical datasets examined by DeMiguel et al., none of the fourteen optimized portfolio strategies consistently outperformed the 1/N benchmark in Sharpe ratio, certainty-equivalent return, and turnover.

I see two advantages to using this benchmark in the Mexican market:

1. A hypothesis to be tested in this research is that the relatively concentrated structure of the Mexican equity universe may exacerbate covariance-estimation problems.

2. For parameters calibrated by DeMiguel et al. to U.S. equity data, they estimate that a 25-asset mean-variance portfolio may require more than 3,000 months of observations before outperforming 1/N. The Mexican market study spans a substantially shorter historical sample than the estimation windows, while also using a time-varying IPC constituent universe. Considering that the IPC index comprises between 35 and 37 assets (depending on the year) using the 1/N strategy (which relies solely on the number of assets and total capital) allows us to determine whether the strategies being tested truly deliver strong market performance when analyzed out-of-sample.